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Editorial Board
eLife Sciences Publications

Dear Editors,

I am submitting the manuscript "Spectral Exponents of the Twelve-Lead ECG Reveal the Anatomy of Cardiac Conduction Disorders and a Bifurcation Between Aging and Disease" for consideration as a Research Article at eLife.

Summary. I apply IRASA—a method from computational neuroscience—to extract the spectral exponent β from the aperiodic component of 412,730 twelve-lead ECGs across three continents. The central finding is a cross-population invariant: despite factor-of-two variation in absolute β-values across equipment, the twelve-lead β-vector distinguishes complete left from complete right bundle branch block with AUC $= 0.982$ on the German discovery cohort, AUC $= 0.982$ on a Chinese replication cohort, and AUC $= 0.979$ on a Brazilian replication cohort. This geometry is determined by the anatomy of the conduction system and is therefore invariant across populations.

Why eLife. I believe this work aligns with eLife's mission in three ways:

1. Rigorous replication over novelty. The core claim rests on identical replication across three independent datasets spanning three continents, recording systems, and populations—rather than on a single-dataset discovery. The analysis also includes an honest null result: β does not independently predict mortality (adjusted HR $= 1.02$, $p = 0.83$), which I report because it calibrates what β is (a diagnostic marker of conduction anatomy) and what it is not (a prognostic biomarker).
2. Interpretability over performance. The 12-lead β-vector provides an interpretable, physics-based characterization of conduction system integrity—each subtype produces a spatial fingerprint that maps directly onto lesion anatomy. This complements rather than competes with deep learning approaches that achieve marginal AUC improvements at the cost of interpretability.
3. Open science. All three datasets are publicly available. The analysis code is fully open: a Kaggle notebook for reproducibility (https://www.kaggle.com/code/theodorspiro/the-heartbeat-at-the-e and a GitHub repository (https://github.com/mool32/ecg-spectral-exponents). A preprint is available on bioRxiv [DOI to be added].

Calibration of claims. The manuscript explicitly separates replicated findings (diagnostic landscape, spatial fingerprints, CLBBB vs. CRBBB AUC) from single-dataset observations (aging bifurcation, subclinical detection) and null results (mortality, absolute β equipment-dependence). The aging trajectory did not replicate externally and is presented as a PTB-XL observation requiring independent validation.

Suggested reviewers:

- Bradley Voytek (UC San Diego) — expertise in $1/f$ neural dynamics and IRASA methodology
- Ary L. Goldberger (Harvard / Beth Israel) — originator of the "loss of complexity" framework in cardiac physiology
- Nils Strodthoff (Fraunhofer HHI) — creator of the PTB-XL benchmark and deep learning ECG analysis
- Antonio H. Ribeiro (Uppsala University) — creator of the CODE-15% dataset and DNN-based ECG age prediction
- Hagen Schmidt (Radboud University) — recent work on IRASA-based ECG spectral slopes and aging

Thank you for considering this manuscript.

Sincerely,

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